

Lab#01 Generation and Observation of Analog and Digital Modulation Techniques

Objective

- To generate analog (AM and FM) and digital (PWM and FSK) modulated signals using a function generator.
- To observe and compare the waveforms using an oscilloscope.
- To examine the influence of modulation parameters on signal characteristics.

Troubleshooting Guidelines

- Ensure that the **function generator output is enabled** (Output ON).
- Confirm that the oscilloscope probe is connected to the **MAIN OUTPUT** of the function generator.
- Verify that the probe is securely connected at both the function generator and oscilloscope.
- If the signal output looks incorrect/noisy try using a **different oscilloscope channel** (e.g., Channel 2 instead of Channel 1).
- If available, replace the probe or BNC cable with another one to rule out faulty connections.
- Check that the function generator is configured for the correct **modulation mode** (AM, FM, PWM, or FSK) and that all parameters are set correctly.
- Adjust the **Volts/Div** and **Time/Div on oscilloscope** settings so that the waveform is clearly visible, with approximately **3–4 cycles** displayed on the screen.

Introduction

Modulation is the process of varying one or more characteristics of a high-frequency carrier signal in accordance with a lower-frequency information signal. It enables efficient transmission of information over communication channels while improving signal quality, transmission range, and bandwidth utilization.

In this experiment, a function generator is used to generate four common modulation techniques: Amplitude Modulation (AM), Frequency Modulation (FM), Pulse Width Modulation (PWM), and Frequency Shift Keying (FSK). The resulting waveforms are observed and analyzed using an oscilloscope. By varying selected modulation parameters while keeping other parameters constant, students investigate how each parameter influences the characteristics of the modulated signal.

Theory

Amplitude Modulation (AM)

Amplitude Modulation (AM) is an analog modulation technique in which the amplitude of a high-frequency carrier signal varies in proportion to the instantaneous amplitude of the modulating signal, while the carrier frequency remains constant. Increasing the modulation depth increases the variation in the carrier envelope. Excessive modulation depth can lead to over-modulation and waveform distortion, as shown in Figure 1.

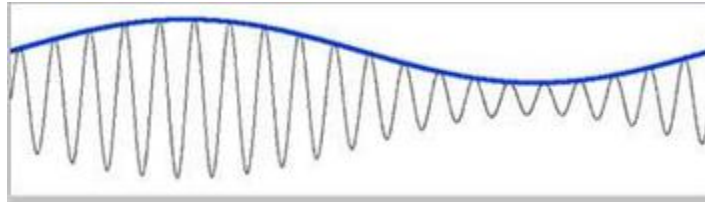


Figure 1: AM waveform

Frequency Modulation (FM)

Frequency Modulation (FM) is an analog modulation technique in which the instantaneous frequency of the carrier signal varies according to the amplitude of the modulating signal, while the carrier amplitude remains constant. The extent of frequency variation is determined by the frequency deviation. Increasing the deviation increases the bandwidth occupied by the FM signal, as shown in Figure 2.

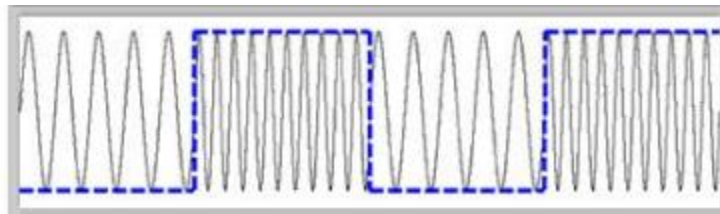


Figure 2: FM waveform

Pulse Width Modulation (PWM)

Pulse Width Modulation (PWM) is a digital modulation technique in which the width (or duty cycle) of a sequence of constant-amplitude pulses is varied according to the modulating signal. The pulse frequency and amplitude remain constant while the pulse width changes. PWM is widely used in motor speed control, power electronics, LED dimming, and digital control systems, as shown in Figure 3.



Figure 3: PWM waveform

Frequency Shift Keying (FSK)

Frequency Shift Keying (FSK) is a digital modulation technique in which digital information is represented by switching the carrier between two discrete frequencies. One frequency typically represents binary '1' (mark) while the other represents binary '0' (space). The amplitude remains constant throughout transmission, making FSK more resistant to amplitude noise than amplitude-based modulation techniques, as shown in Figure 4.

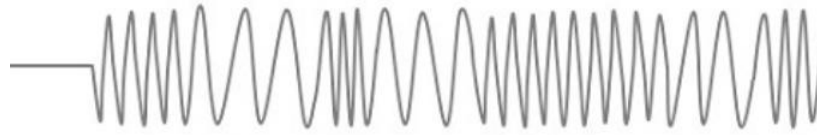


Figure 4: FSK waveform

Experimental Work Apparatus Required

The following apparatus are required to perform this experiment.

No.	Apparatus	Specifications	Quantity
1	Function Generator	AM/FM/PWM/FSK, 1 MHz	1
2	Oscilloscope	0-30 MHz	1

Steps:

1. Set the **carrier frequency and amplitude** required for the experiment.
2. Press the **MOD** button and select the required modulation type and parameters.
3. Verify that the required settings are displayed, for example:

Type: PWM

Source: INT

Shape: Square

PWM Duty Cycle: xx %

PWM Frequency: xx Hz

(The displayed parameters will vary depending on the selected modulation type. For AM/FM/FSK, select Sine as the modulation shape.)

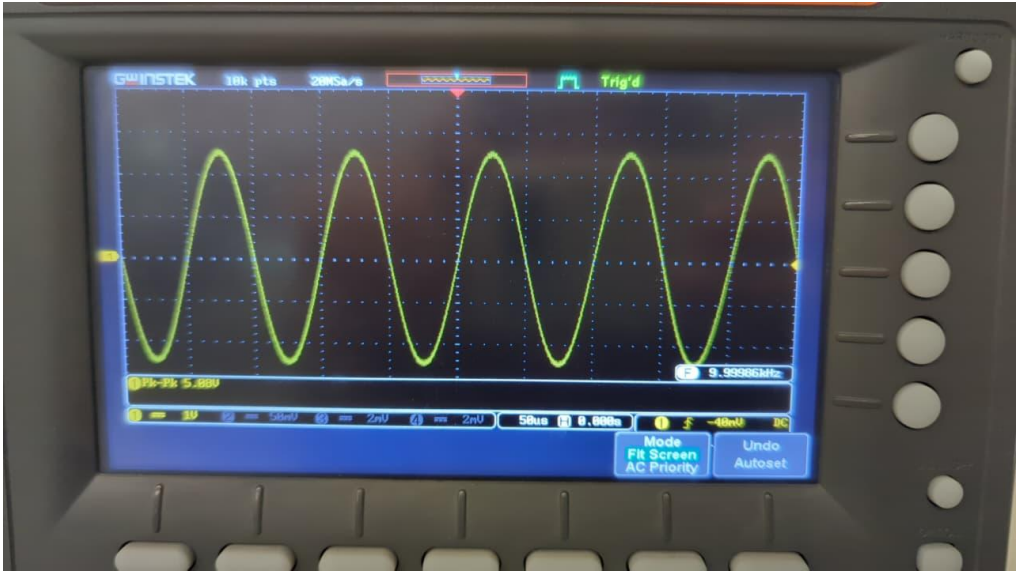
4. Enable the **MOD** function and observe the modulated output signal at the **MAIN** terminal of the function generator using an oscilloscope.
5. If required, observe the **modulating signal** at the **MOD** terminal of the function generator using an oscilloscope.

6. Disable the **MOD** function and observe the unmodulated carrier signal at the **MAIN** terminal using an oscilloscope.

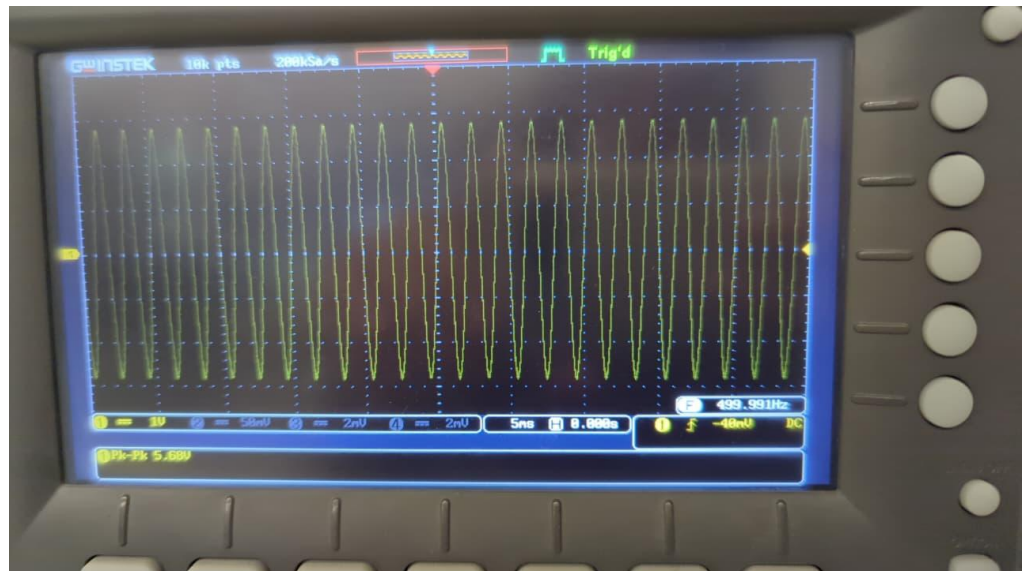
Task 1: Amplitude Modulation (AM)

Part A: Effect of Modulation Depth

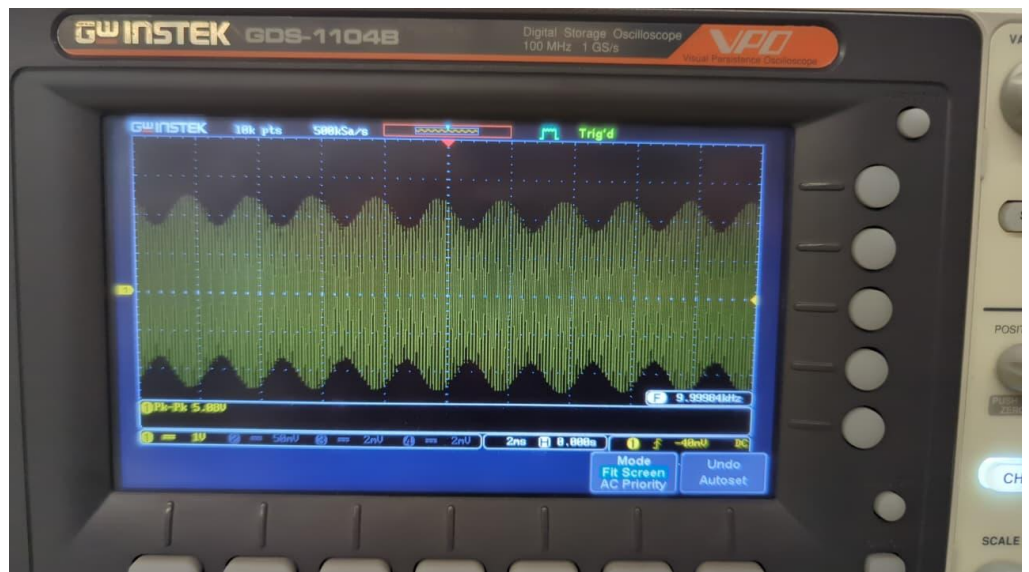
Configure the function generator to generate an AM signal. Set the carrier frequency to 10 kHz and the carrier amplitude to 5 Vpp. Keep the modulating frequency constant at 500 Hz and vary the modulation depth through the values in the table. Observe the resulting waveform on the oscilloscope and note the changes in the signal envelope. Capture and attach the oscilloscope waveform for each modulation depth.

Modulation Depth	Oscilloscope Waveform
Carrier signal	

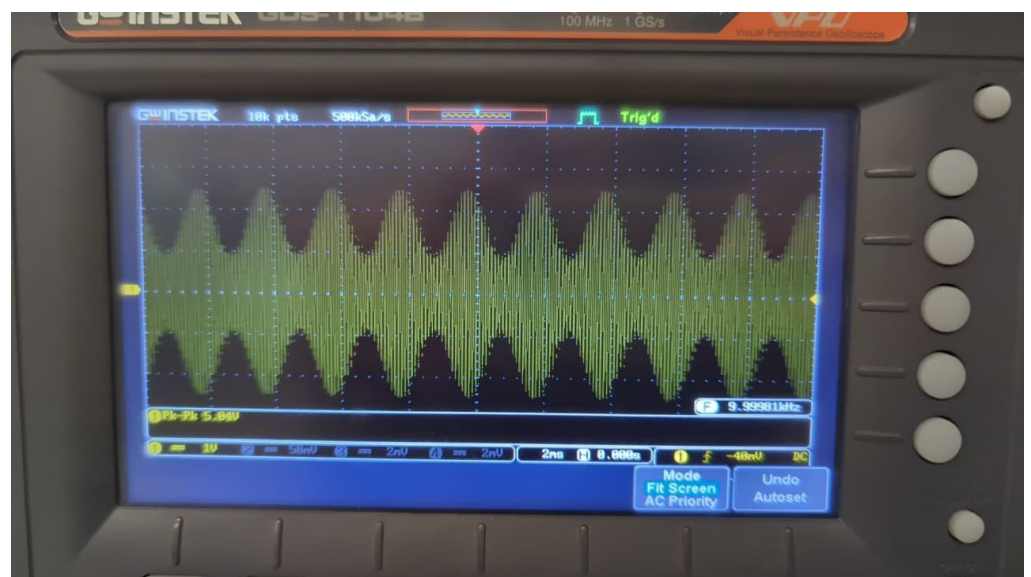
Modulating signal



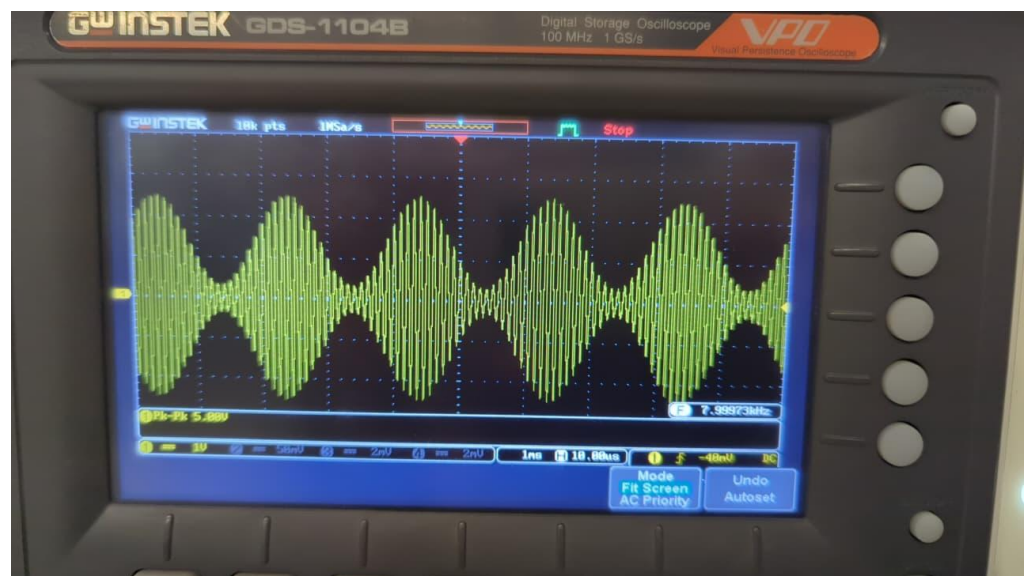
20%
Expected output:
Frequency:10k Hz
Amplitude:5 Vpp



50%



80%

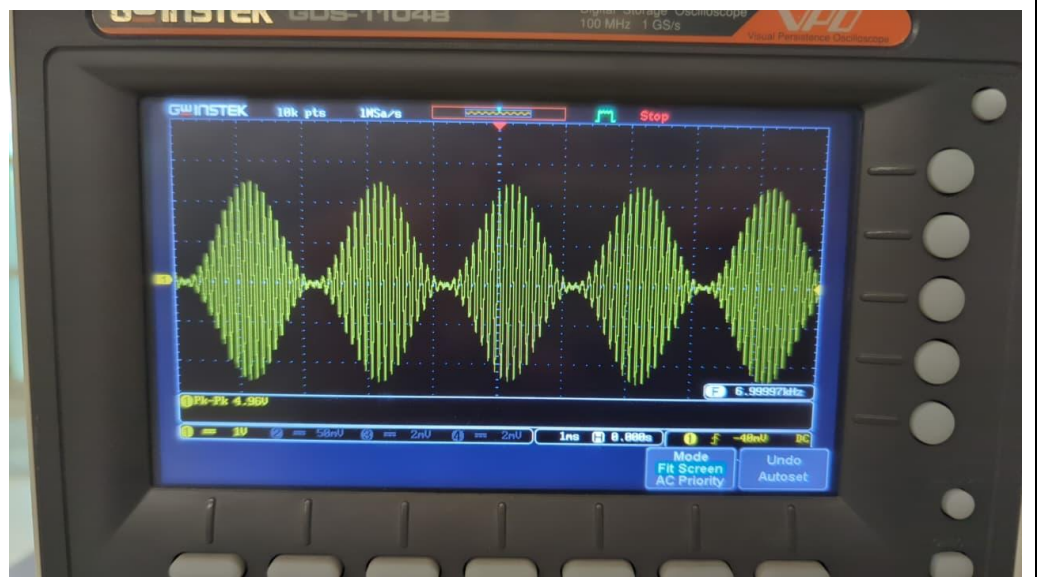


100%

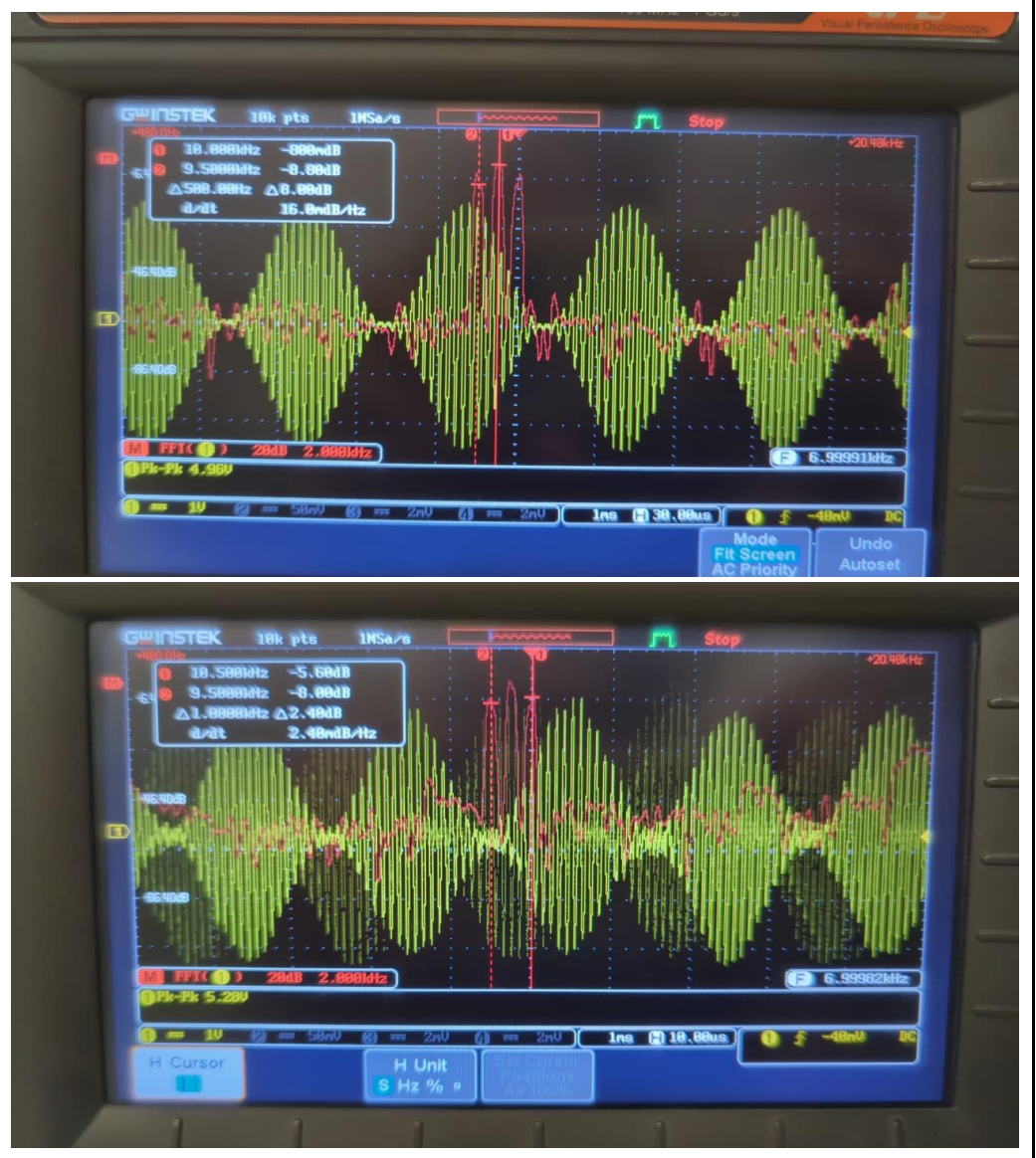
Expected output:

Frequency: 10k Hz

Amplitude: 5 Vpp



Frequency Spectrum
of AM waveform at
100% Modulation
depth



120%

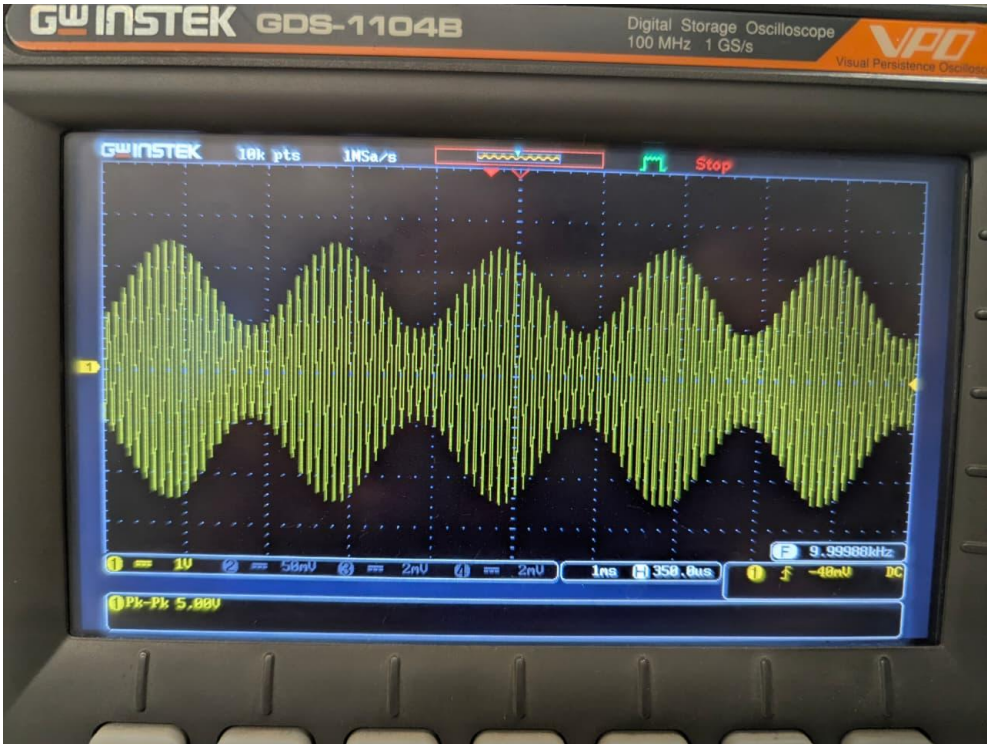


What changes do you observe in the amplitude and envelope of the AM waveform when the modulation depth is varied?

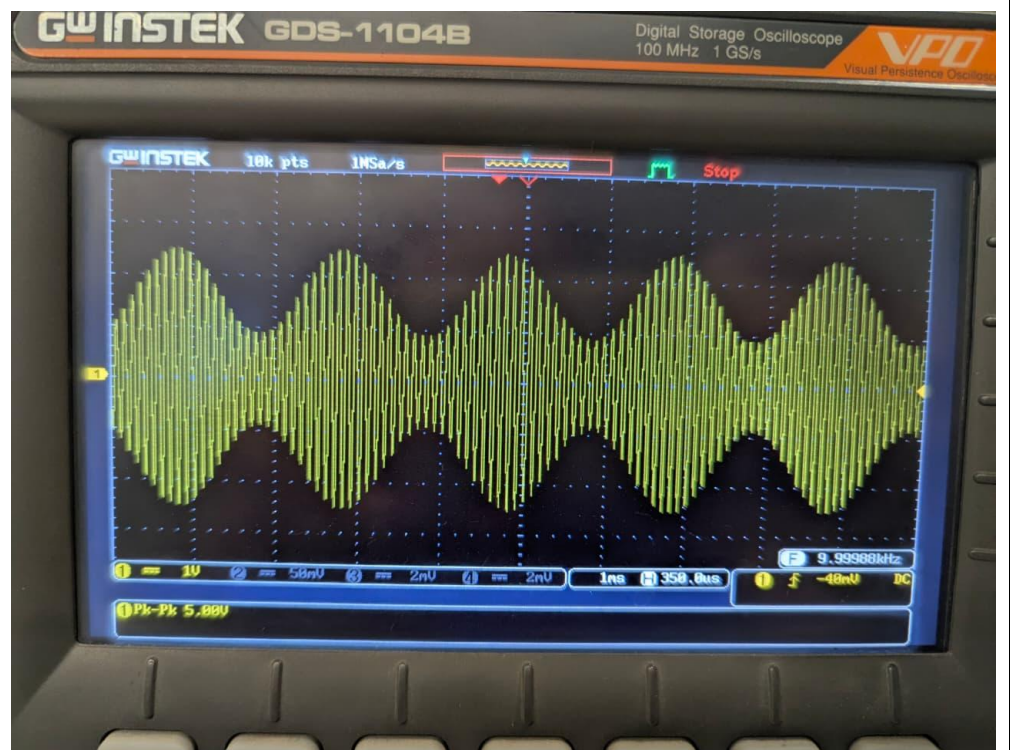
As the modulation depth increases, the overall amplitude of the carrier wave stays constant, but the amplitude of its top and bottom envelope varies more widely to match the shape of the modulating signal. At low percentages like 20% or 50%, the envelope undergoes mild variations and stays far above the zero line. At 100% modulation depth, the envelope reaches its maximum variation where the minimum amplitude touches zero. If the modulation depth goes beyond 100% (over-modulation), the envelope becomes distorted and crosses over itself because the signal is temporarily cut off.

Part B: Effect of Modulating Frequency

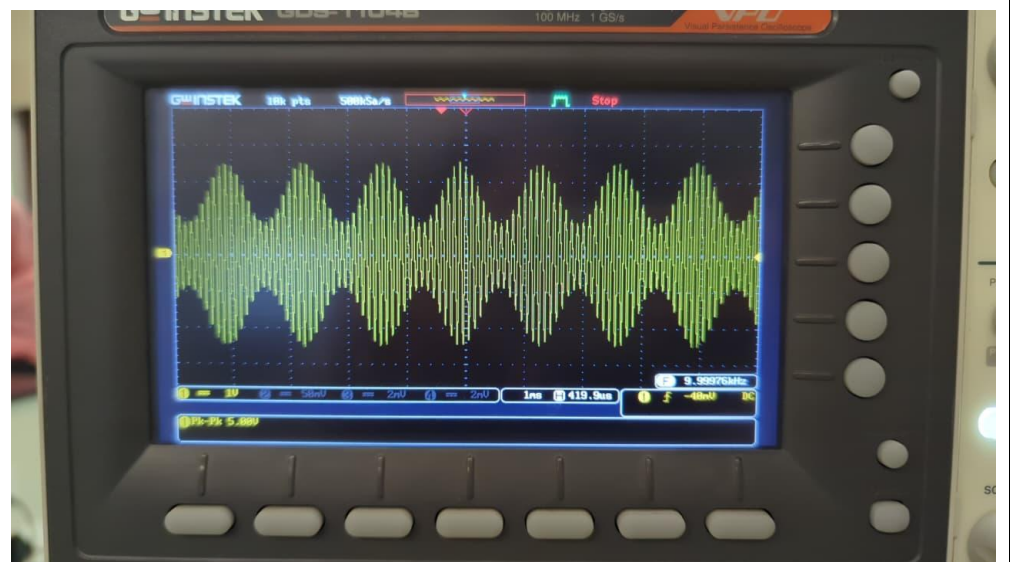
Keep the carrier frequency at 10 kHz, the carrier amplitude at 5 Vpp, and the modulation depth at 50%. Vary the modulating frequency through the values in the table. Observe the resulting waveform on the oscilloscope and note the changes in the modulation envelope. Capture and attach the oscilloscope waveform for each modulating frequency.

Modulating Frequency	Oscilloscope Waveform
250 Hz Expected output: Frequency:10k Hz Amplitude:5 Vpp	

500 Hz



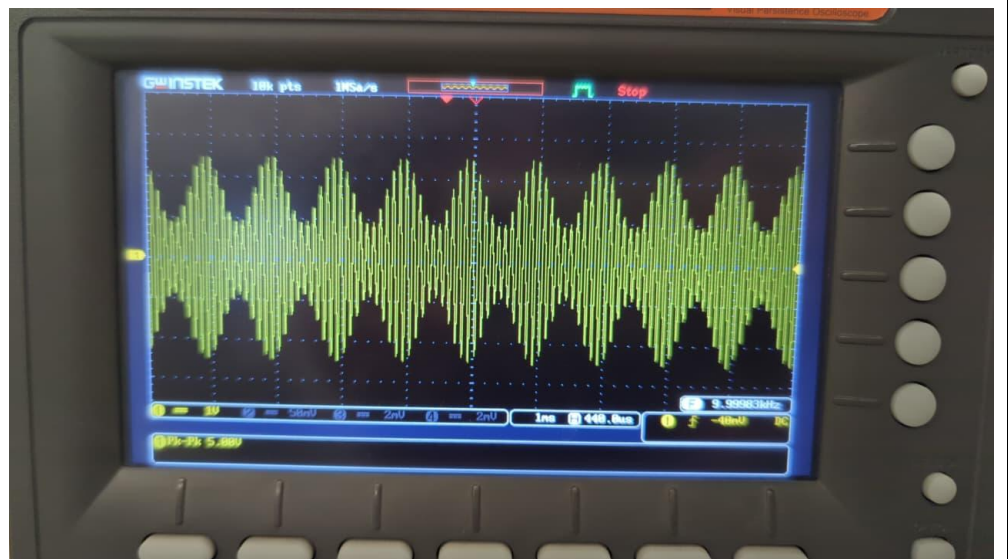
750 Hz



Frequency Spectrum of
FM waveform at 750Hz
modulating frequency



1 kHz
Expected output:
Frequency:10k Hz
Amplitude:5 Vpp



What changes do you observe in the AM waveform when the modulating frequency is varied?

Varying the modulating frequency changes the rate at which the envelope expands and contracts without altering the overall signal amplitude or modulation depth. Increasing the frequency causes the envelope to oscillate faster, packing more modulating cycles across the oscilloscope screen. Conversely, decreasing the frequency slows down the envelope variations, resulting in wider, fewer cycles displayed.

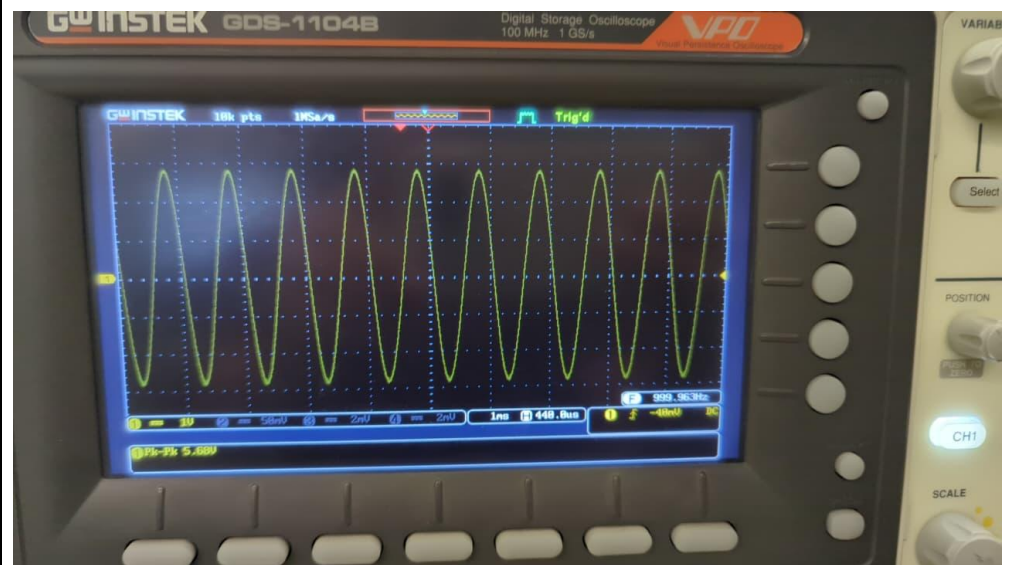
Task 2: Frequency Modulation (FM)

Part A: Effect of Frequency Deviation

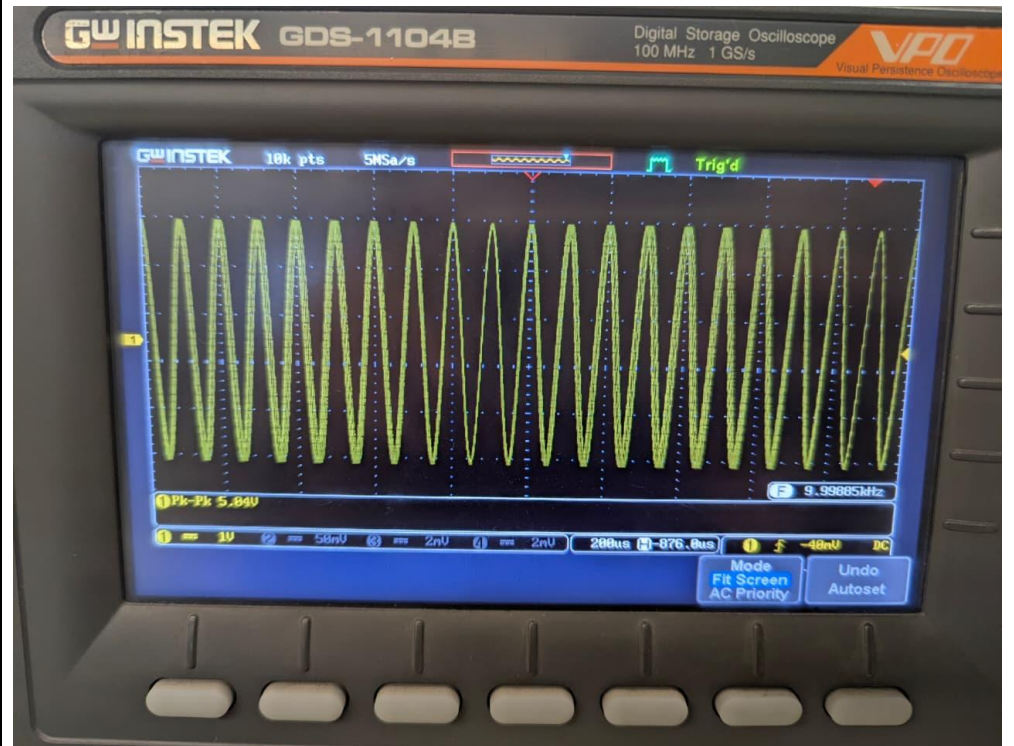
Configure the function generator to generate an FM signal. Set the carrier frequency to 10 kHz and the carrier amplitude to 5 Vpp. Keep the modulating frequency constant at 1 kHz and vary the frequency deviation through the values in the table. Observe the waveform on the oscilloscope and note the changes in the instantaneous frequency of the carrier. Capture and attach the corresponding oscilloscope waveform for each frequency deviation.

Frequency Deviation	Oscilloscope Waveform
Carrier signal	 A photograph of a GWINSTEK oscilloscope screen displaying a carrier signal. The screen shows a green sine wave with a peak-to-peak amplitude of 5.00V. The frequency is set to 9.93804 MHz. The time scale is 200us. The screen also displays various settings like 18k pts, 5MSa/s, and a Trig'd indicator.

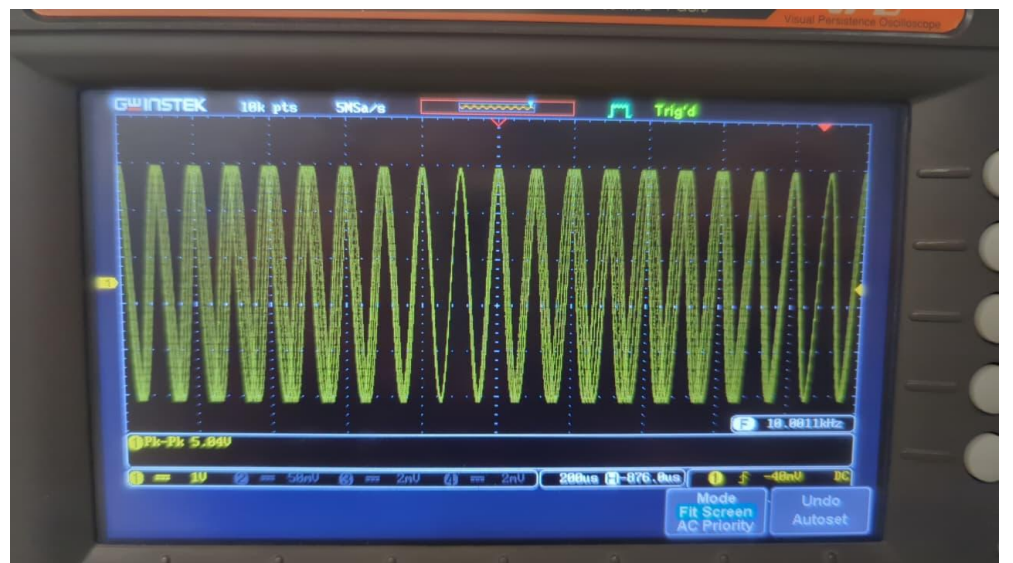
Modulating signal



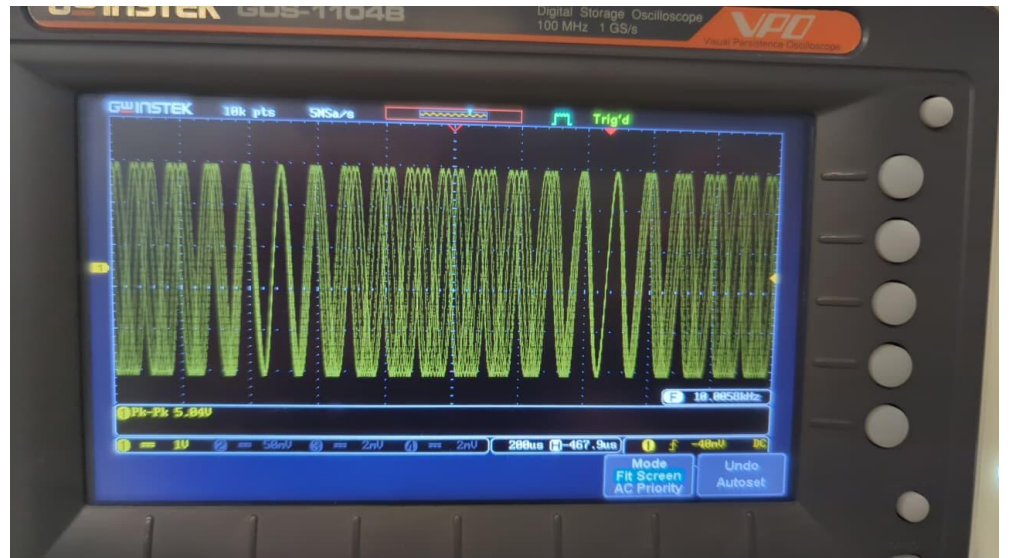
250 Hz
Expected output:
Frequency:10k Hz
Amplitude:5 Vpp



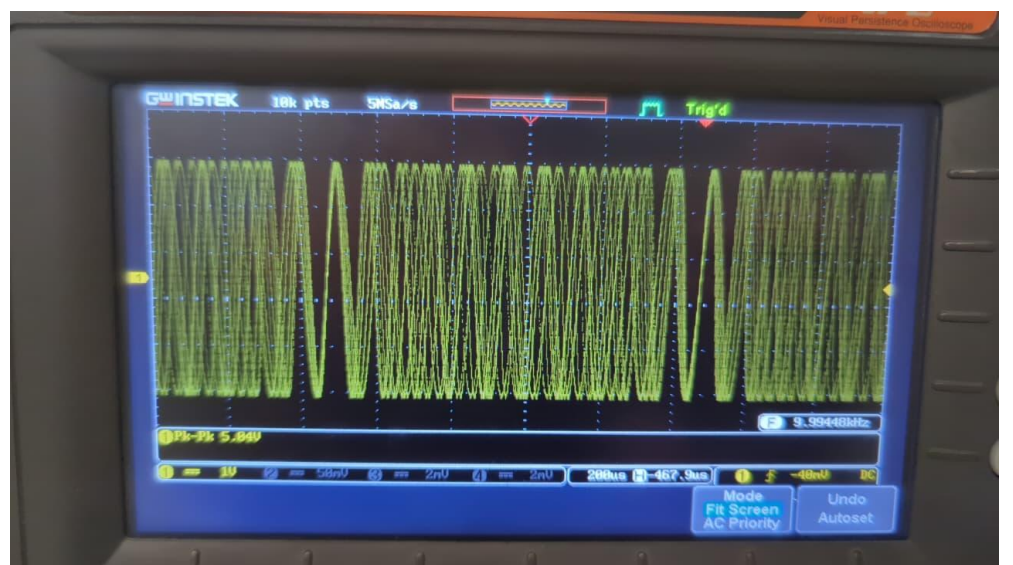
500 Hz



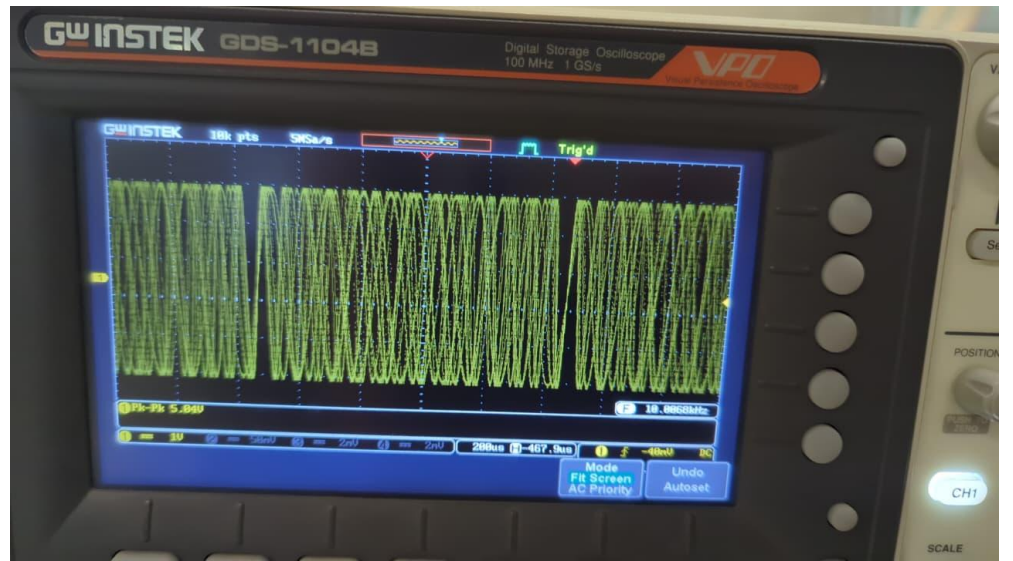
1 kHz



2 kHz



5 kHz
Expected output:
Frequency:10k Hz
Amplitude:5 Vpp



FFT OF FM

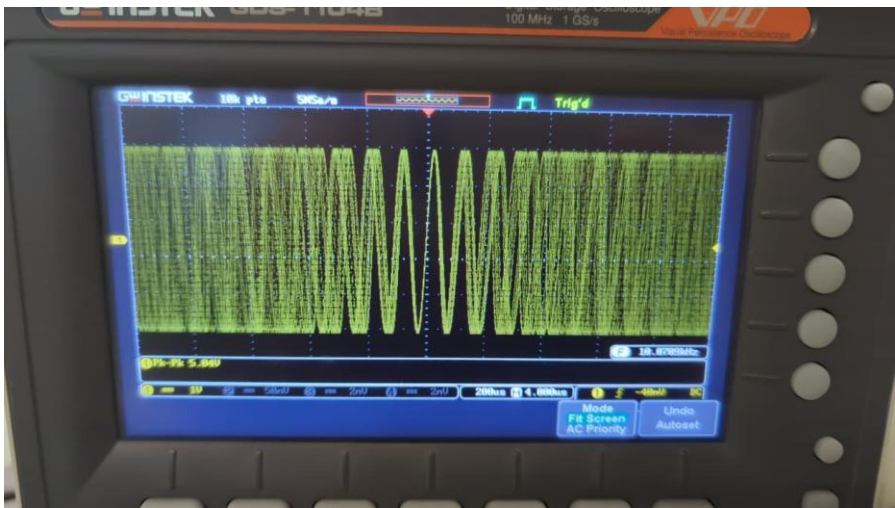
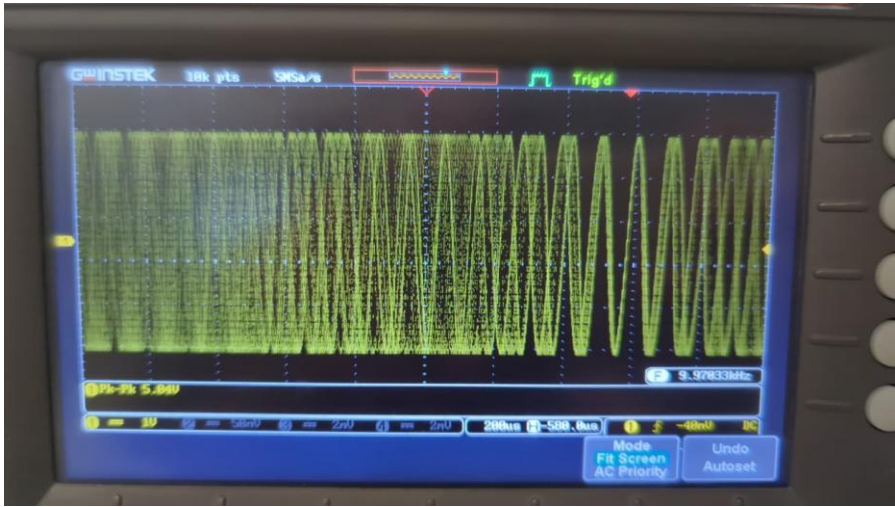


What changes do you observe in the spacing and instantaneous frequency of the FM waveform when the frequency deviation is varied?

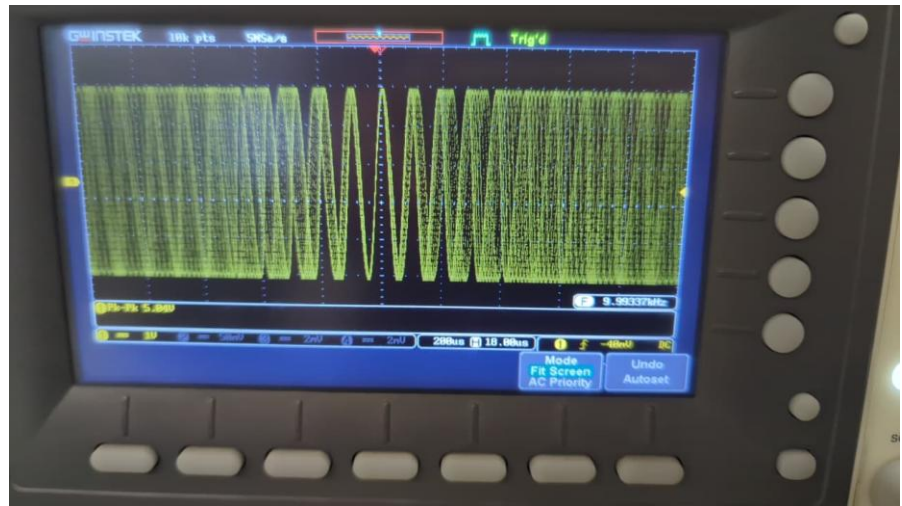
Increasing the frequency deviation expands the range between the maximum and minimum instantaneous frequencies of the FM wave, causing the spacing between waveform cycles to vary more drastically. At higher deviations, the cycles become tightly packed together during peak frequency regions and noticeably spread far apart during low frequency regions. At lower deviations, the cycle spacing remains nearly uniform and harder to distinguish visually. Throughout all changes in frequency deviation and cycle spacing, the overall peak-to-peak amplitude of the FM signal remains completely constant.

Part B: Effect of Modulating Frequency

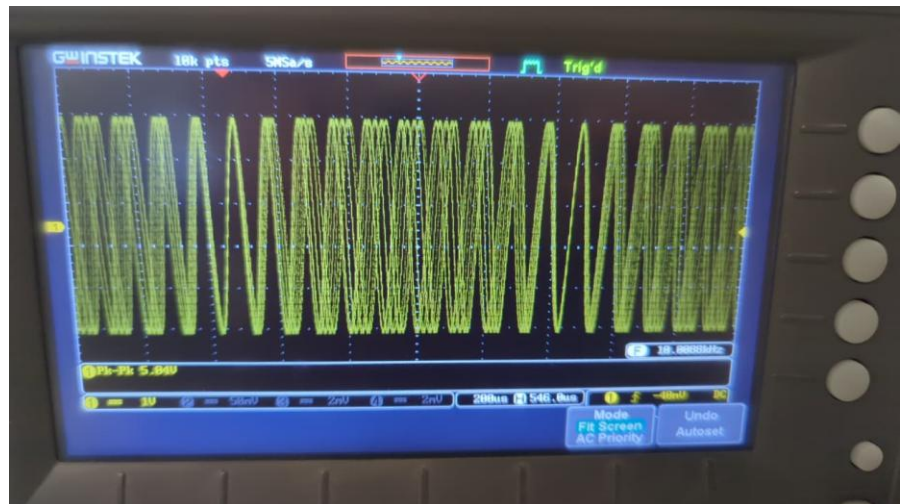
Keep the carrier frequency at 10 kHz, the carrier amplitude at 5 Vpp, and the frequency deviation at 1 kHz. Vary the modulating frequency through the values in the table. Observe the waveform on the oscilloscope and note the changes in the rate at which the carrier frequency varies. Capture and attach the corresponding oscilloscope waveform for each modulating frequency.

Modulating Frequency	Oscilloscope Waveform
100 Hz Expected output: Frequency: 10 kHz Amplitude: 5 Vpp	 The oscilloscope screen shows a green waveform representing a 10 kHz carrier wave modulated by a 100 Hz signal. The carrier wave's frequency varies sinusoidally at the 100 Hz rate. The screen displays '10.000000' at the bottom right, indicating the carrier frequency. The top status bar shows '100 MHz' and '1 GS/s'. The bottom status bar shows '5.000V' and '200ns'.
250 Hz	 The oscilloscope screen shows a green waveform representing a 10 kHz carrier wave modulated by a 250 Hz signal. The carrier wave's frequency varies sinusoidally at the 250 Hz rate. The screen displays '9.92833MHz' at the bottom right, indicating the carrier frequency. The top status bar shows '100 MHz' and '1 GS/s'. The bottom status bar shows '5.000V' and '200ns'.

500 Hz



1 kHz

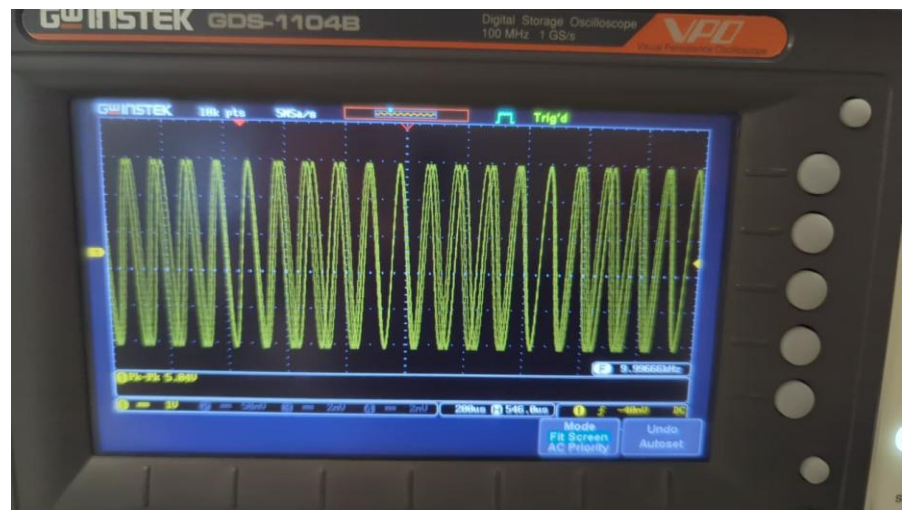


2 kHz

Expected output:

Frequency: 10 kHz

Amplitude: 5 Vpp



What changes do you observe in the FM waveform when the modulating frequency is varied?

Varying the modulating frequency alters how quickly the FM waveform alternates between its high-frequency and low-frequency regions, without changing the total range of frequency expansion or peak signal amplitude. Increasing the modulating frequency causes the rate of frequency variations to speed up, resulting in more rapid transitions between compressed and stretched cycles across the oscilloscope screen. Conversely, decreasing the modulating frequency slows down this switching rate, producing wider bands of high- and low-frequency cycles that change more gradually over time.

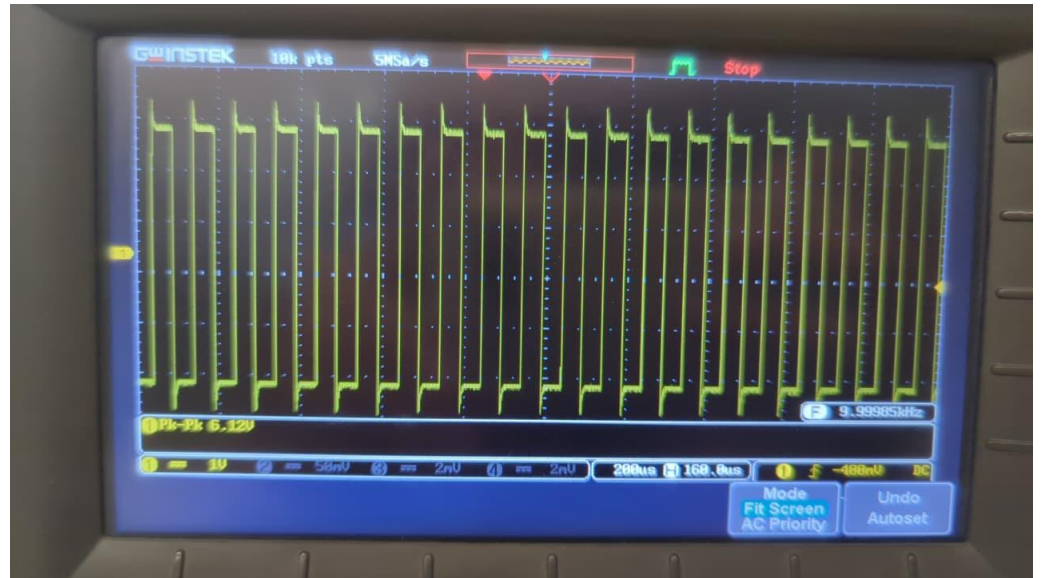
Task 3: Pulse Width Modulation (PWM)

Part A: Effect of Duty Cycle

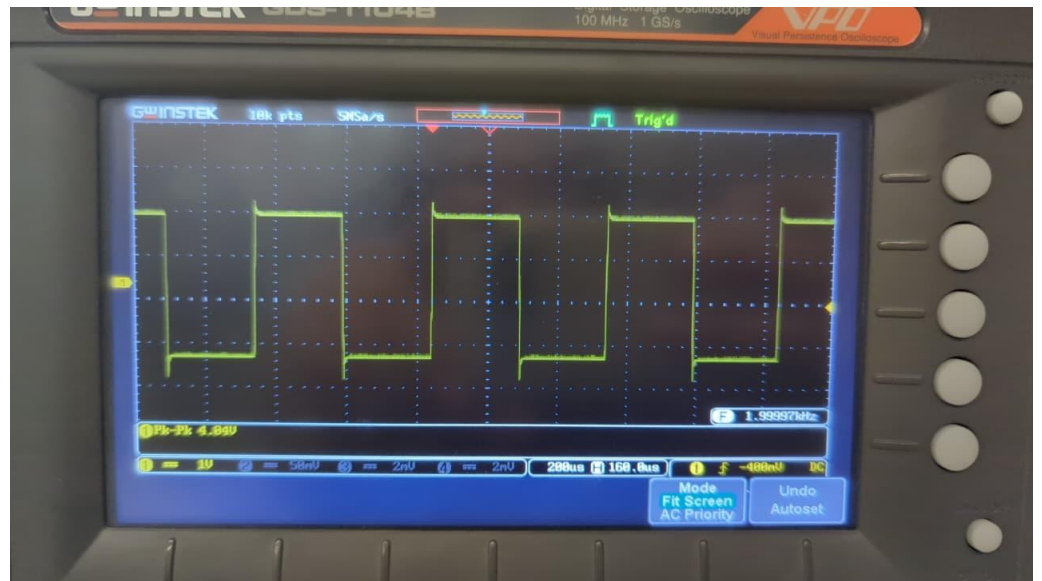
Configure the function generator to generate a PWM signal. Set the carrier frequency to 10 kHz and the signal amplitude to 5 Vpp. Keep the PWM frequency constant at 100 Hz and vary the duty cycle through the values in the table. Observe the waveform on the oscilloscope and note the changes in the pulse width. Capture and attach the corresponding oscilloscope waveform for each duty cycle.

Duty Cycle	Oscilloscope Waveform
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Carrier signal

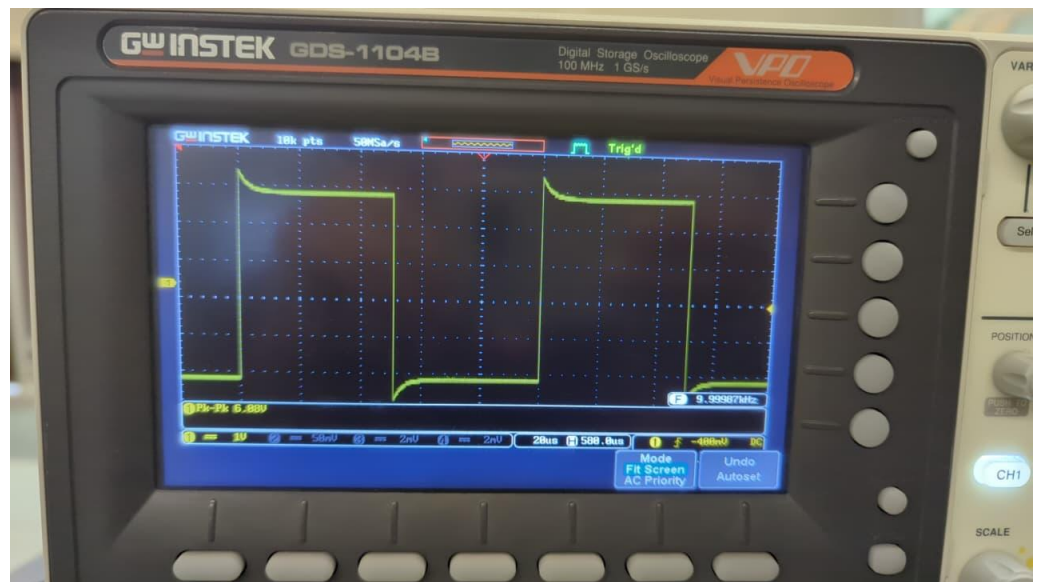


Modulating signal



0%

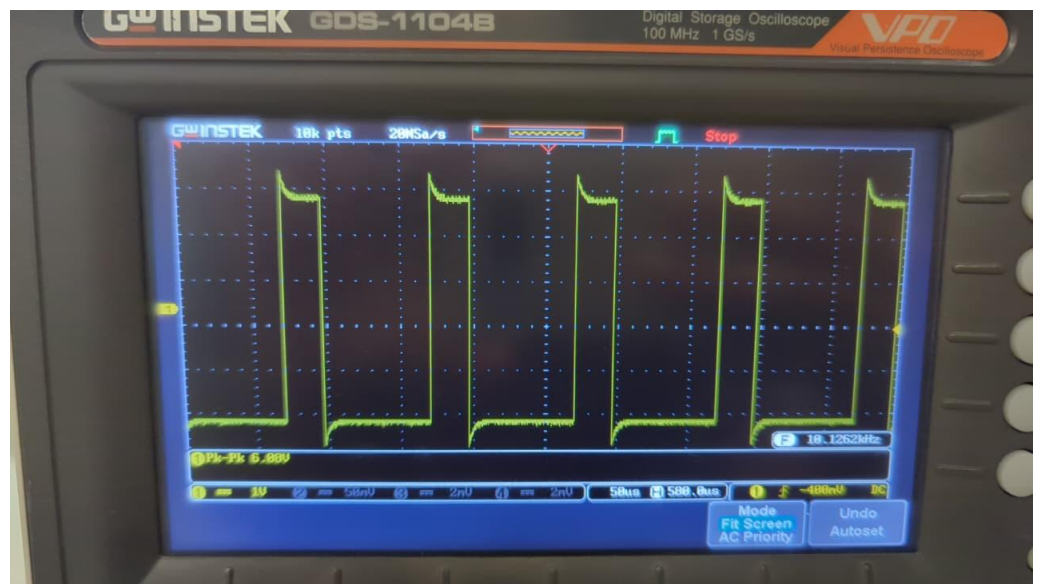
Expected output:
Frequency: 10k Hz
Amplitude: 5 Vpp



20%



50%



80%



100%

Expected output:

Frequency: 10k Hz

Amplitude: 5 Vpp



What changes do you observe in the pulse width of the PWM waveform when the duty cycle is varied?

Varying the duty cycle directly alters the width of the positive pulses while keeping the overall period, pulse frequency, and signal amplitude completely constant. Increasing the duty cycle expands the active high time (pulse width) of each cycle relative to its low time, covering up to 100% where the waveform becomes a continuous high DC level. Conversely, decreasing the duty cycle narrows the active high time, shortening the pulse width down toward 0% where the signal remains flat at the low voltage level.

Part B: Effect of PWM Frequency

Keep the carrier frequency at 10 kHz, the signal amplitude at 5 Vpp, and the duty cycle constant at 50%. Vary the PWM frequency through the values in the table. Observe the waveform on the oscilloscope and note the changes in the rate at which the pulse width varies. Capture and attach the corresponding oscilloscope waveform for each PWM frequency.

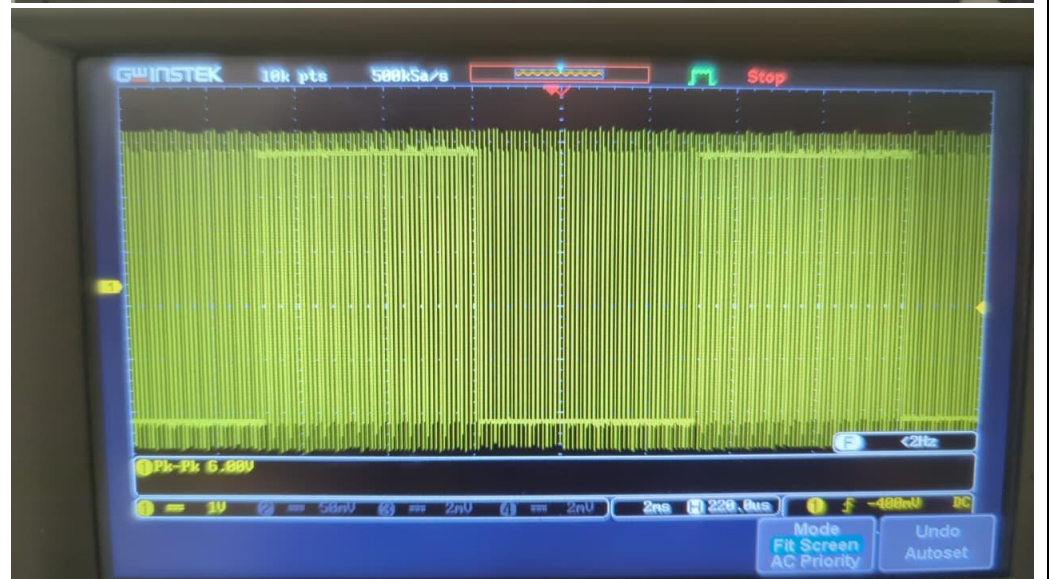
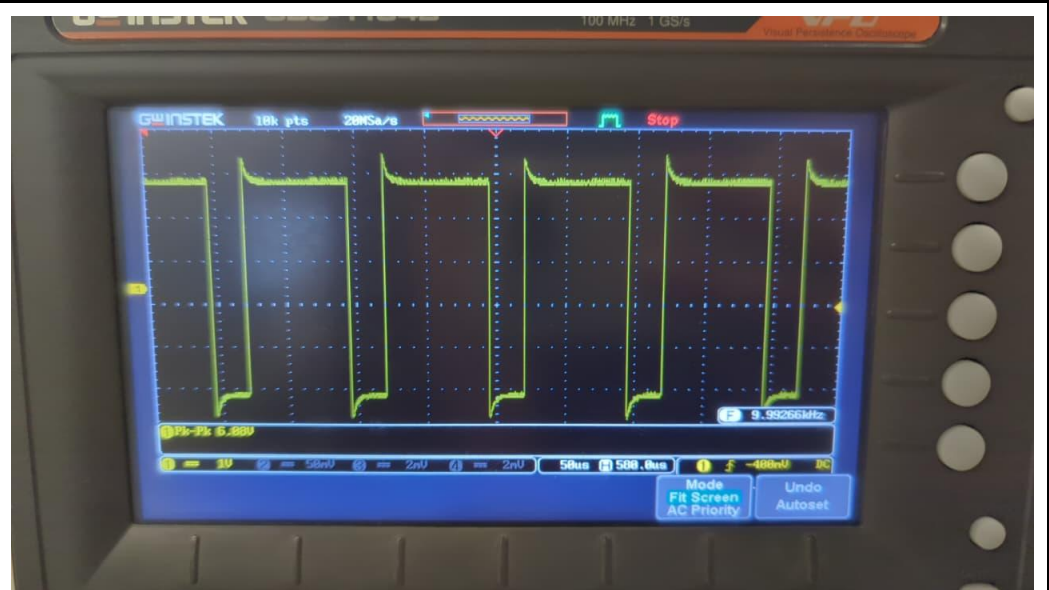
PWM Frequency	Oscilloscope Waveform
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100 Hz

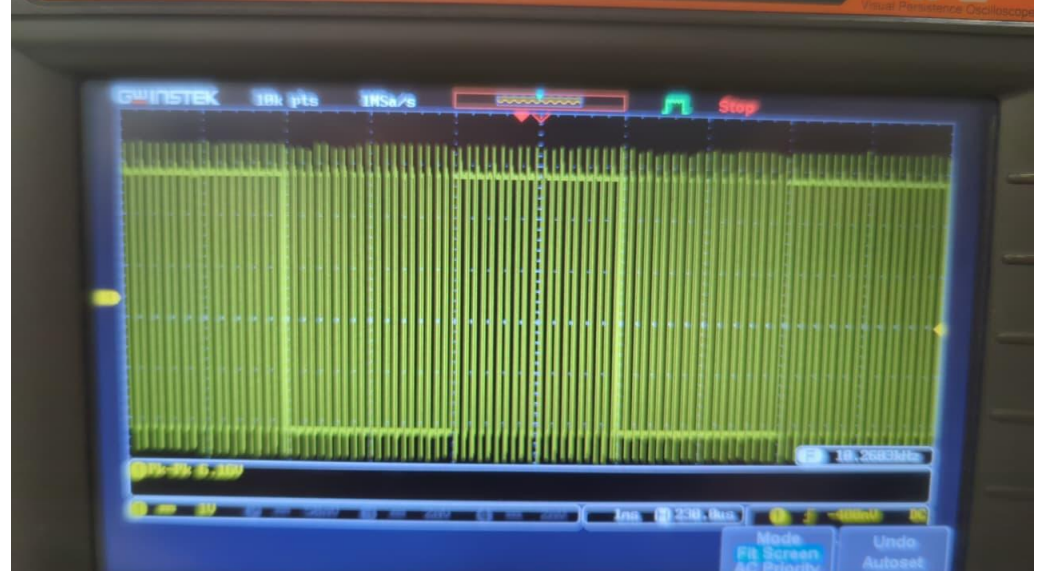
Expected output:

Frequency: 10k Hz

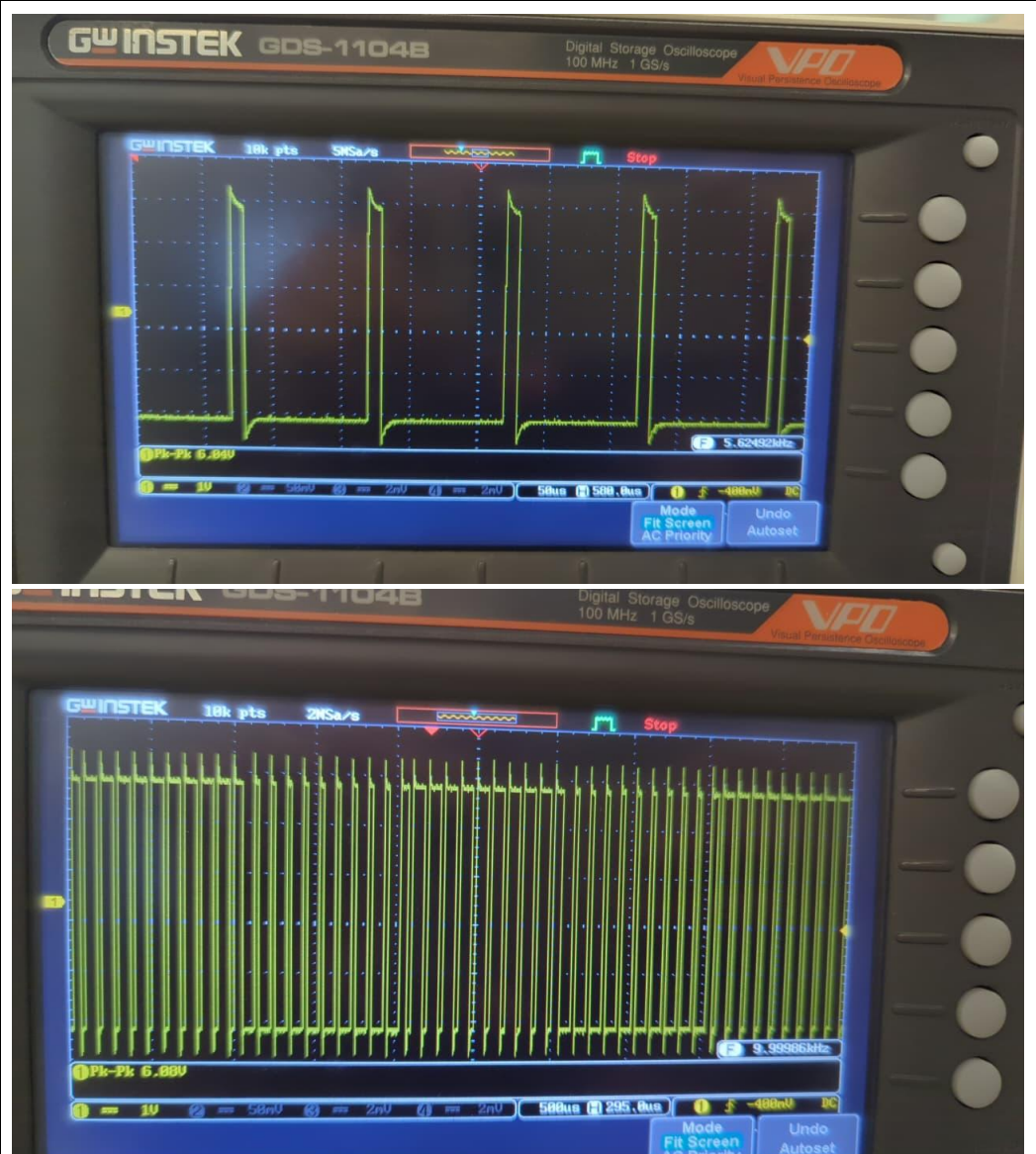
Amplitude: 5 Vpp



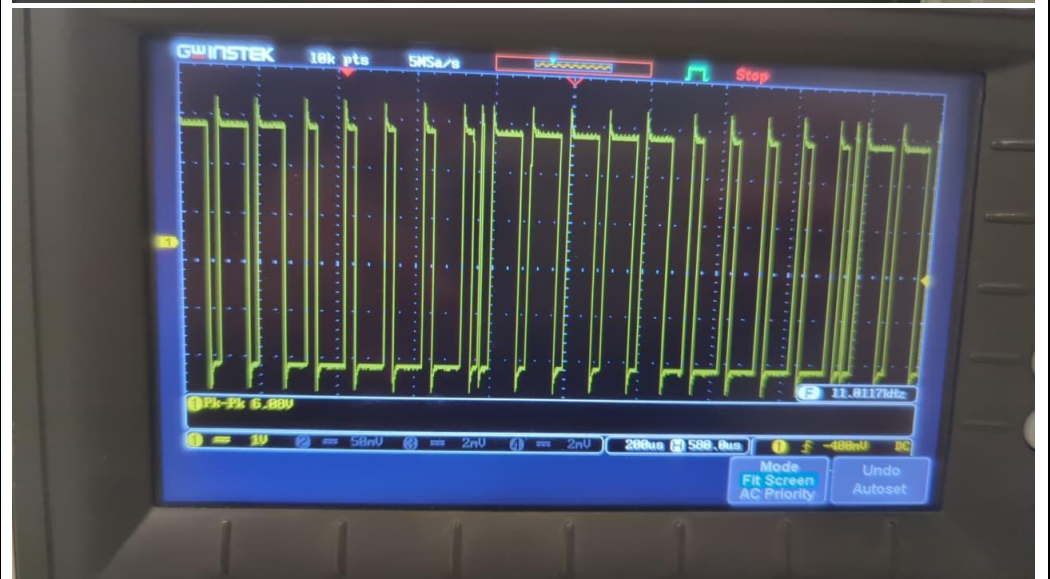
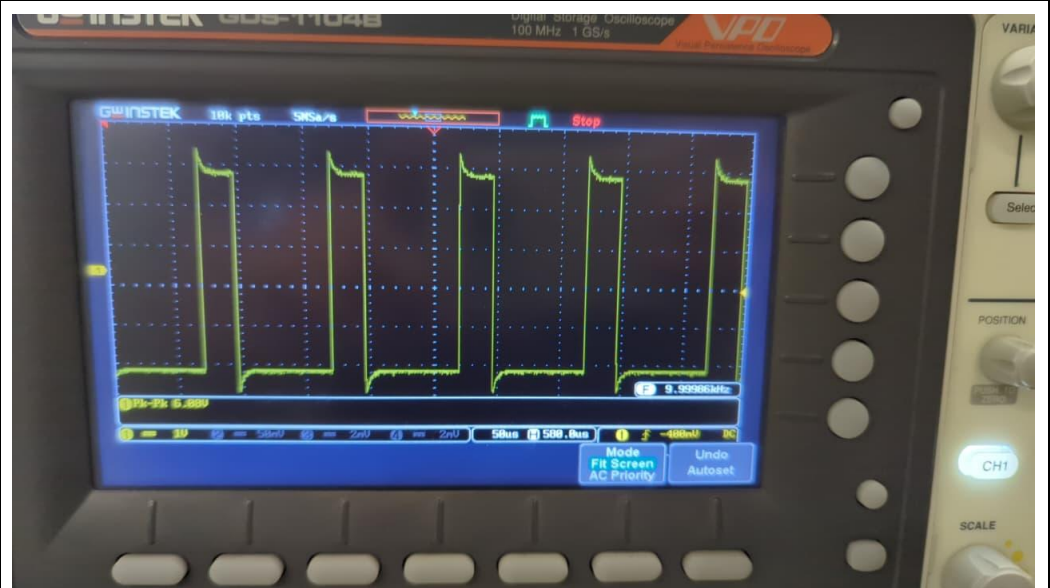
250 Hz



500 Hz



1 kHz

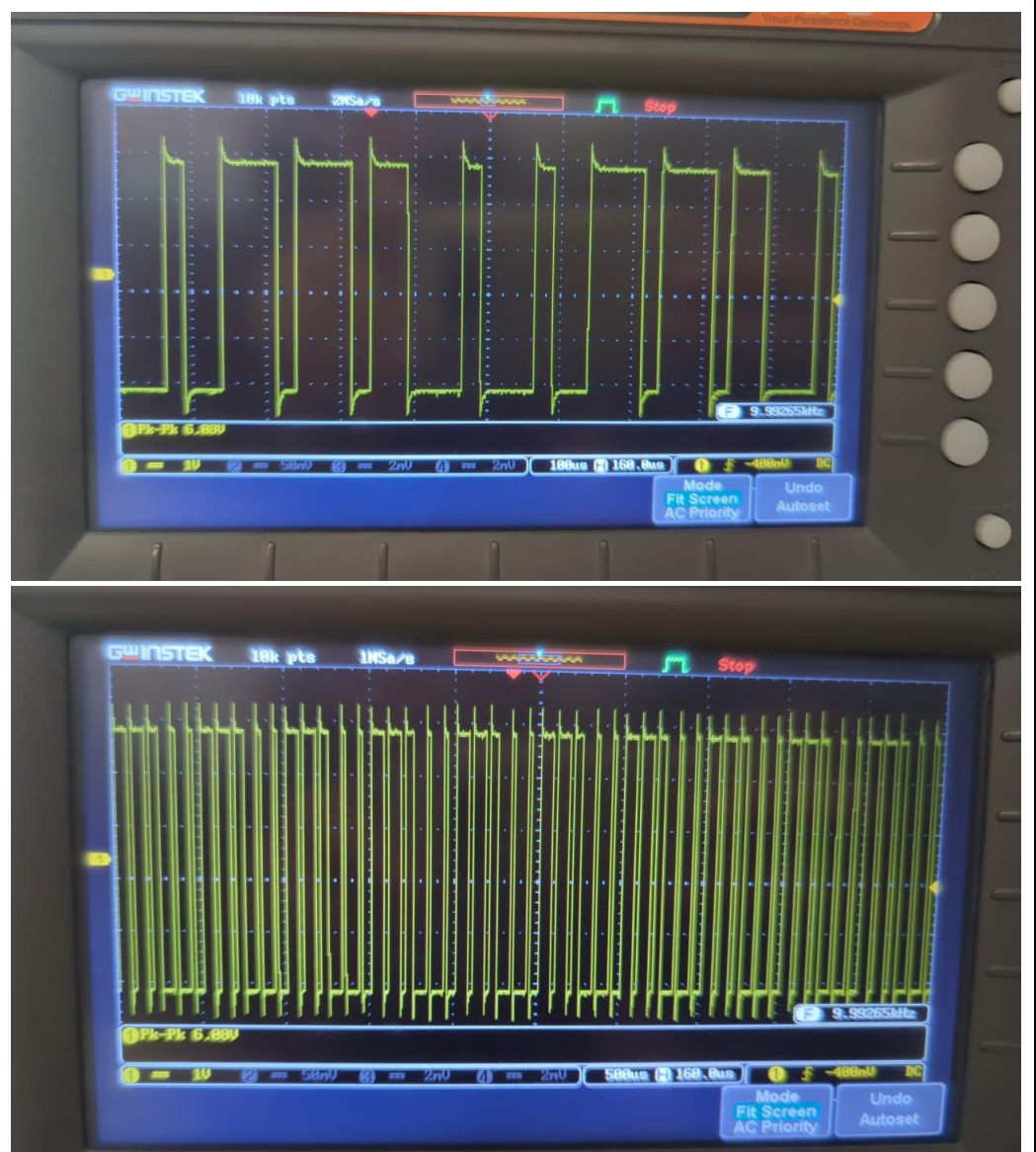


2 kHz

Expected output:

Frequency: 10 kHz

Amplitude: 5 Vpp



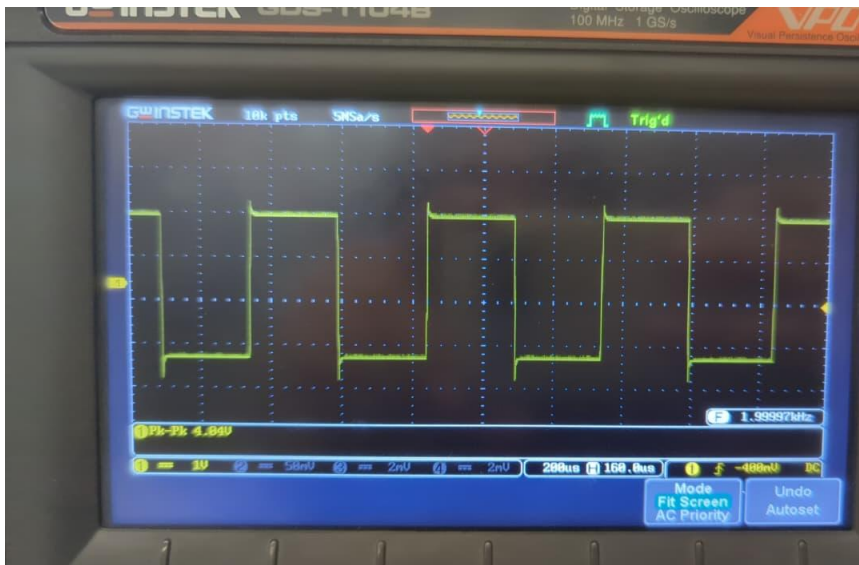
What changes do you observe in the PWM waveform when the PWM frequency is varied?

Varying the PWM frequency changes the rate at which the pulses repeat over time while keeping the signal amplitude and active duty cycle percentage fixed. Increasing the PWM frequency shortens the total period of each cycle, causing the fixed duty cycle ratio (high time vs. low time) to repeat much faster and packing more narrow pulse cycles across the oscilloscope screen. Conversely, lowering the PWM frequency lengthens the cycle period, causing the pulses to stretch out and repeat at a slower rate.

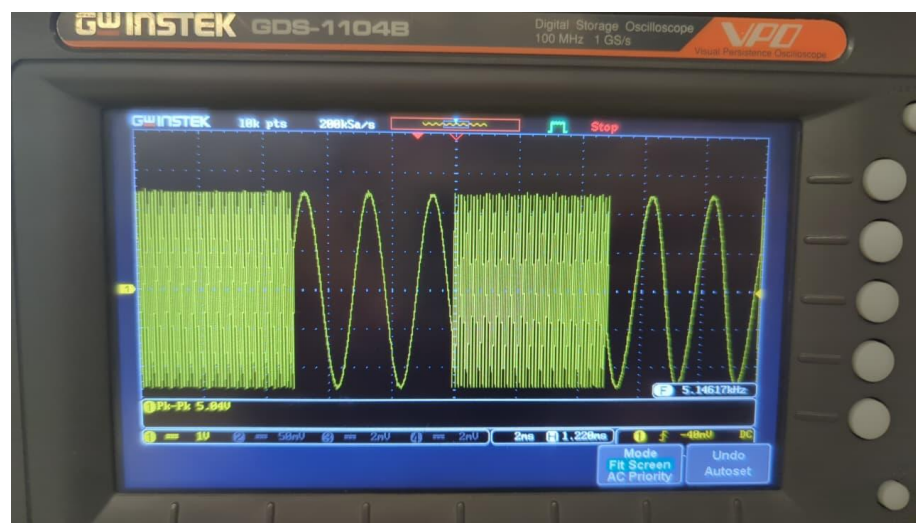
Task 4: Frequency Shift Keying (FSK)

Part A: Effect of Hop Frequency

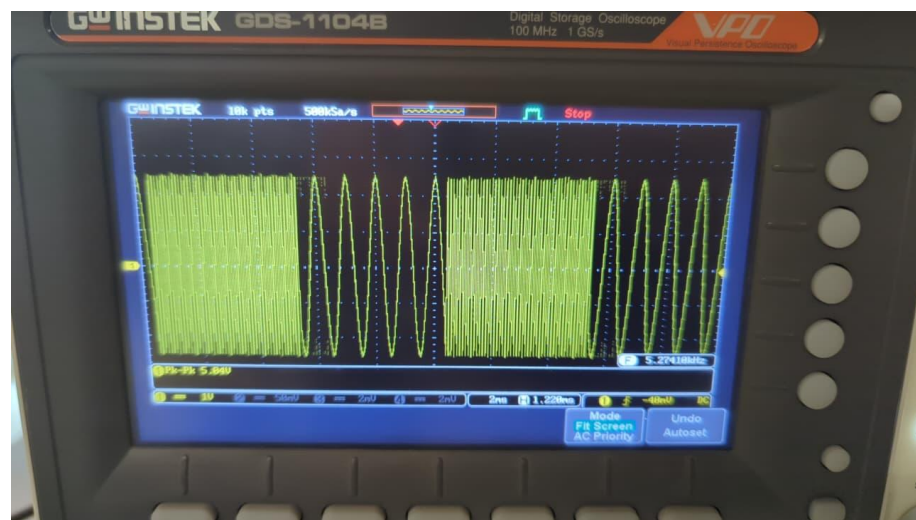
Configure the function generator to generate an FSK signal. Set the carrier frequency to 10 kHz and the carrier amplitude to 5 Vpp. Keep the FSK rate constant at 100 Hz and vary the hop frequency through the values in the table. Observe the waveform on the oscilloscope and note the changes in the separation between the transmitted frequencies. Capture and attach the corresponding oscilloscope waveform for each hop frequency.

Hop Frequency	Oscilloscope Waveform
Carrier signal	
Modulating signal	

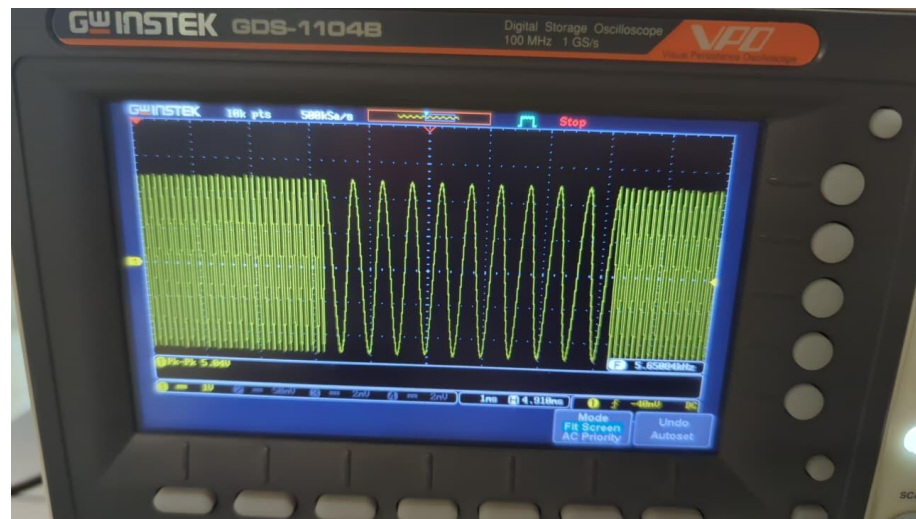
500 Hz
Expected output:
Frequency:10k
Hz
Amplitude:5 Vpp



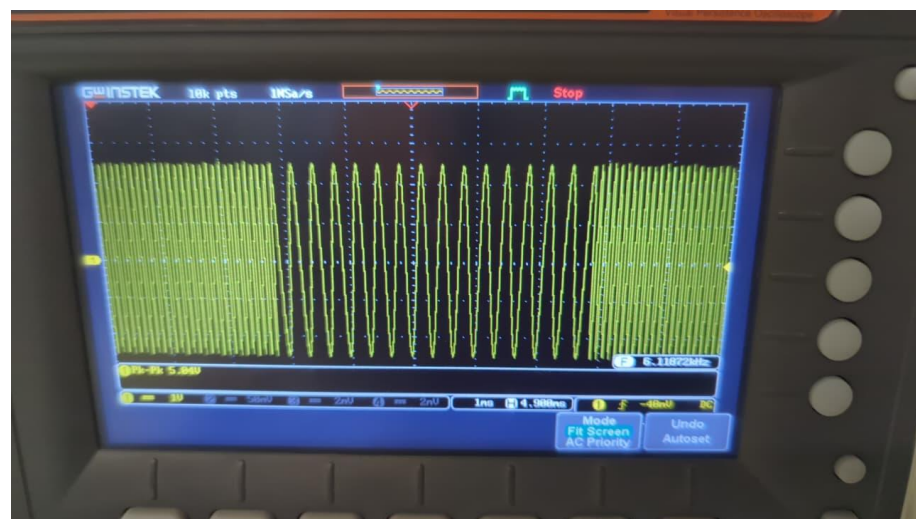
1 kHz



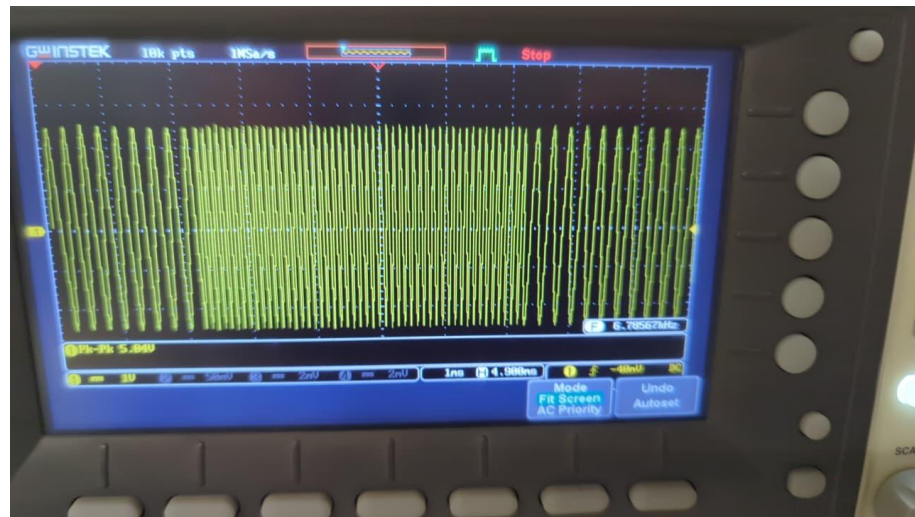
2 kHz



3 kHz



4 kHz
Expected output:
Frequency:10k
Hz
Amplitude:5 Vpp



What changes do you observe in the separation between the two frequencies of the FSK waveform when the hop frequency is varied?

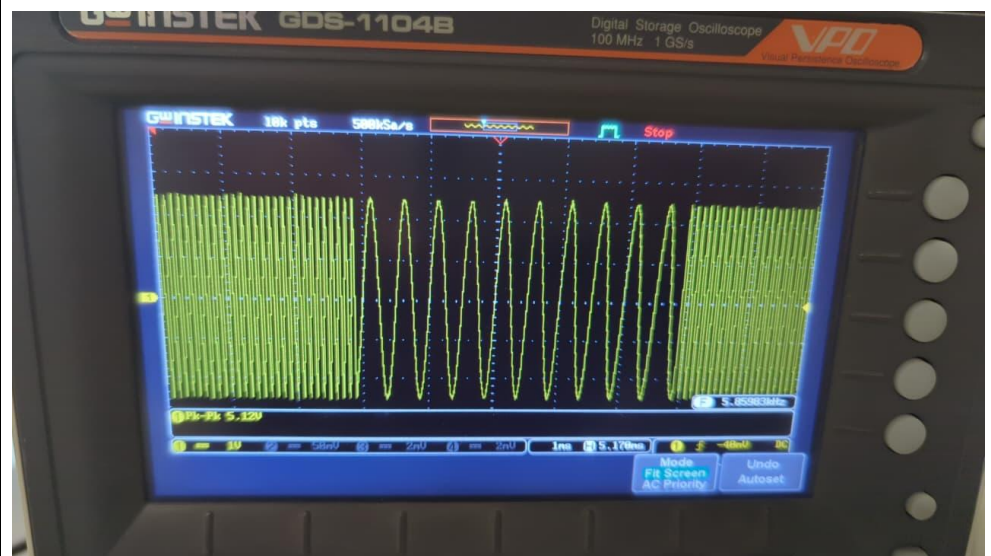
Varying the hop frequency directly changes the difference (or distance) between the carrier frequency and the second transmitted frequency, while keeping the FSK switching rate and signal amplitude constant. Increasing the hop frequency creates a larger frequency shift, making the visual separation on the oscilloscope between the closely spaced high-frequency cycles and the wider low-frequency cycles much more distinct. Conversely, reducing the hop frequency brings the two frequencies closer together, narrowing their separation and making the transition between the two states harder to distinguish visually.

Part B: Effect of FSK Rate

Keep the carrier frequency at 10 kHz, the carrier amplitude at 5 Vpp, and the hop frequency constant at 2 kHz. Vary the FSK rate through the values in the table. Observe the waveform on the oscilloscope and note the changes in the switching rate between the two frequencies. Capture and attach the corresponding oscilloscope waveform for each FSK rate.

FSK Rate	Oscilloscope Waveform
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100 Hz
Expected output:
Frequency:10k
Hz
Amplitude:5 Vpp



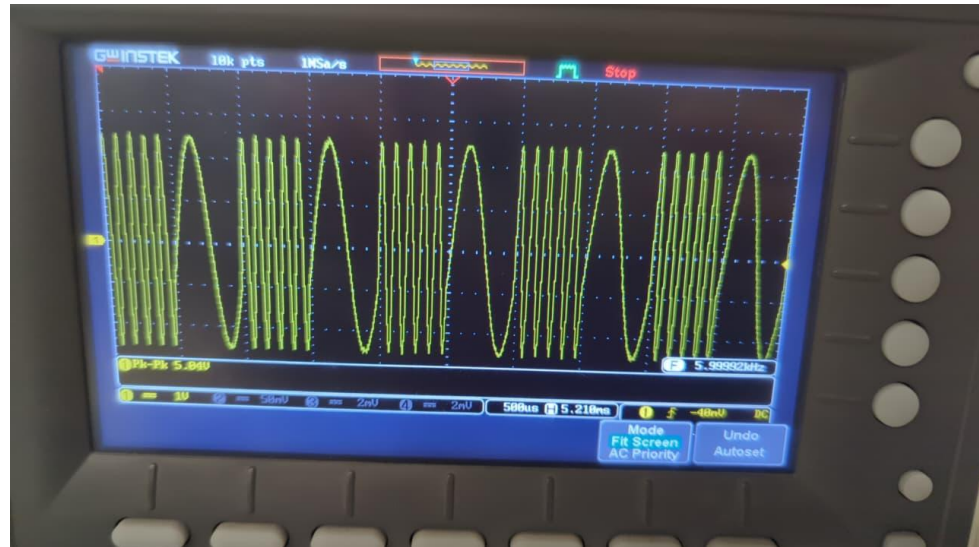
250 Hz



500 Hz



1 kHz



2 kHz
Expected output:
Frequency: 10k
Hz
Amplitude: 5 Vpp



What changes do you observe in the switching rate between the two frequencies when the FSK rate is varied?

Varying the FSK rate changes how quickly the signal alternates between the two set frequencies, while keeping the specific frequency values and peak amplitude constant. Increasing the FSK rate speeds up the transition rate between the high-frequency and low-frequency regions, packing more frequency shifts across the oscilloscope display. Conversely, lowering the FSK rate slows down the transition rate, keeping the signal on a single frequency state for a longer duration before switching to the other.

DOCX+ 2

Name & ID: _____

Date: _____

EE-322L Analog and Digital Communications Lab

Fall 2026

Habib University

Dhanani School of Science & Engineering



Lab 1: Generation and Observation of Analog and Digital Modulation Techniques

Marks distribution:

		LR1=10	LR3=10	LR4=40	LR5=20	LR6=10	AR4=10
In-Lab	Task 1 (AM)	/10	--	/10	/5	/10	/10
	Task 2 (FM)	--	--	/10	/5		
	Task 3 (PWM)	/10	--	/10	/5		
	Task 4 (FSK)	--	--	/10	/5		
Total Marks		100					

CLO	LDL	Points
2	Psy-4	/90
3	Aff-3	/10
Total		/100

Lab Evaluation Assessment Rubric

#	Assessment Elements	(0<Level 1<=4)	(4<Level 2<=6)	(6<Level 3<=8)	(8<Level 4<=10)
LR1	Correct Instrument Setup	Function generator and oscilloscope are wired incorrectly; output/probe settings not configured. Minimal effort shown.	Instruments are connected but with an untidy or partly incorrect setup (e.g. wrong channel, output not enabled).	Instruments are mostly correctly connected and configured, with only minor setup issues.	Function generator and oscilloscope are correctly connected and configured for each modulation type. Given task completed in due time.
LR3	Troubleshooting	Unable to identify the fault (no output, wrong channel, noisy signal). Minimal effort shown.	Able to identify the fault but unable to resolve it using the troubleshooting guidelines.	Able to identify the fault and partially resolve it.	Able to identify the fault and apply the appropriate troubleshooting steps to correct it.
LR4	Data Collection	Waveforms are incomplete, inaccurate, or missing for most parameter values across the four tasks.	Waveforms are captured for some parameter values but are inaccurate, imprecise, or recorded confusingly.	Waveforms are mostly accurate and captured for nearly all required parameter values. Minor errors present.	Waveforms are accurately and precisely captured for every parameter value in all four tasks (AM, FM, PWM, FSK), matching expected outputs.
LR5	Results & Plots	Captured waveforms contain errors or are poorly presented; missing titles, axes labels, or units.	Most waveforms are OK; some are missing important labels or features.	All waveforms are correctly captured and labeled, but some have minor problems or could be improved.	All waveforms are correctly captured, clearly labeled with axes and units, and include titles/captions for every task.
LR6	Clean-up	Equipment is not powered off. Items left at station and not cleaned.	Equipment mostly not powered off. Many items left at station.	Some equipment not powered off. Some items left at station.	All equipment is powered off. Station left neat and clean.
AR4	*Report Submission	Late submission after 1 week.	Late submission after 4 days and within a week.	Late submission after the lab timing and after 2 days of the due date.	Timely submission of the report and in lab time.

****Report: Report will not be accepted after 1 week of due date***